

Exercise 1 — Linear Algebra

Submission deadline: Monday, June 8th

You are encouraged to use AI models (ChatGPT, Claude, Copilot, etc.) throughout this exercise. Please include your chat logs — with your prompts — as part of your submission. Feel free to reach out with any questions.

Part 1 — Word Embeddings

Write a Python notebook that loads pre-trained word embedding vectors for the following five words: *king*, *queen*, *dog*, *cat*, and *coffee*. You may use an existing library such as **gensim** (which provides access to Word2Vec models).

- What is the dimensionality (length) of the embedding vectors?
- Compute the pairwise distances between all pairs of words. Which pairs are closest? Which are furthest apart?

Part 2 — Solving a Linear System

Consider the following system of three equations:

$$x_1 + x_2 + x_3 = 6$$

$$x_1 - x_2 + x_3 = 2$$

$$x_1 + x_2 - x_3 = 0$$

- Rewrite this system in matrix form $\mathbf{A} @ \mathbf{x} = \mathbf{b}$, where $\mathbf{A}$ is a 3×3 matrix and $\mathbf{x}$, $\mathbf{b}$ are 3×1 column vectors. (Here $@$ denotes matrix multiplication.)
- Assuming $\mathbf{A}$ is invertible, find the solution x_1, x_2, x_3 using $\mathbf{x} = \mathbf{A}^{-1} \mathbf{b}$.

Part 3 — Linearity of Matrix Multiplication

- State the conditions that define a *linear* operation.
- Show that matrix multiplication (as defined in class) satisfies these conditions.

Part 4 — The Polynomial Derivative Matrix

Let $\mathbf{p}$ be a 5×1 column vector whose entries are all equal to 1, and let $\mathbf{D}$ be the 5×5 matrix that represents the derivative operator on polynomials of degree ≤ 4 .

For each expression below, state whether it is well-defined ("kosher") — and if so, compute it:

1. $\mathbf{D} @ \mathbf{p}$
2. $\mathbf{p.T} @ \mathbf{D}$ (*p transposed times D*)
3. $\mathbf{D} @ \mathbf{D}$ (*i.e., $\mathbf{D}^2$*)
4. Compute the scalar $\mathbf{p.T} @ \mathbf{D} @ \mathbf{p}$. Does it have a meaningful interpretation?

5. **General case:** For a vector $\mathbf{p}$ of dimension $n \times 1$ (all ones) and the corresponding $n \times n$ derivative matrix $\mathbf{D}$, what is $\mathbf{p}^T \mathbf{D} \mathbf{p}$ in general?

Good luck!